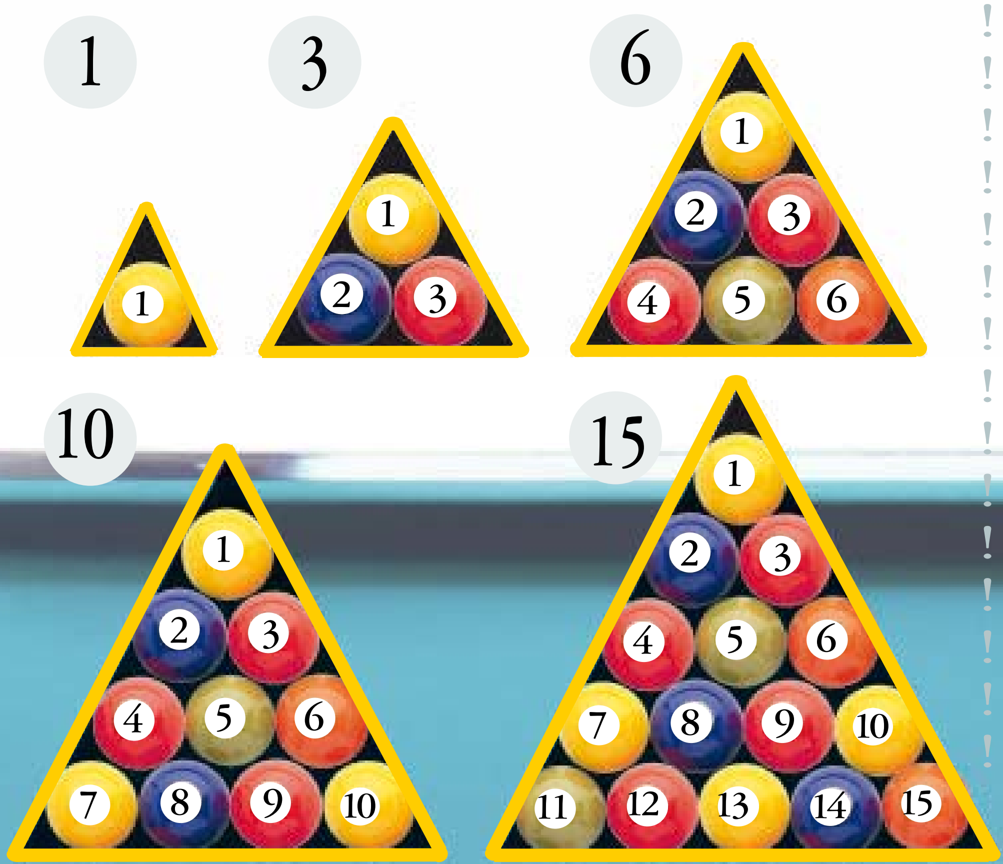
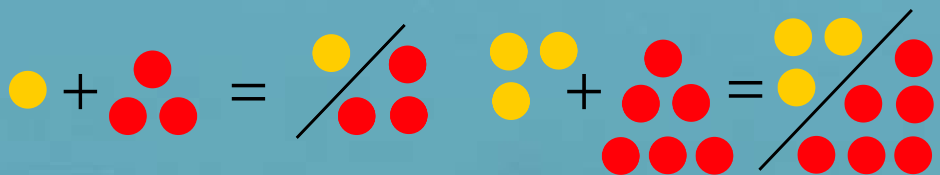
Triangular NUMBERS

Take a pile of marbles and arrange them in triangles. Make each triangle one row bigger than the last, and count the number of marbles in each triangle. You'll end up with another special sequence: ***triangular numbers***.



Squares from triangles

Triangular numbers are full of interesting patterns. Here's one of them: if you add neighbouring triangular numbers together, they always make square numbers. Try it. Mathematicians can prove this mathematically using algebra, but there's an even simpler way to prove it, with pictures:



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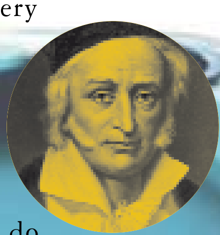
Adding up

A curious fact about triangular numbers is that you can make ***any*** whole number by adding no more than **three triangular numbers**. The number 51, for instance, is 15 + 36. See if you can work out which triangular numbers add up to your age. We know it always works because the rule was ***proved*** 200 years ago by one of the most brilliant mathematicians of all time: Karl Gauss.

Clever or just Gauss work?



Karl Gauss (1777–1855) was a mathematical genius. When he was a schoolboy, his teacher tried to keep the class quiet by telling them to add up every number from 1 to 100. But Gauss stood up within seconds with the right answer: 5050. How did he do it? Like most geniuses, he found a shortcut. He added the first and last numbers (1+100) to get 101. Then he added the second and second-to-last numbers (2+99) to get the same number, 101. He realized he could do this 50 times, so the answer had to be 50 × 101.

Did you know?

Triangular numbers never end in 2, 4, 7, or 9. Every other triangular number is a hexagonal number. If a group of n people shake hands with each other, the total number of handshakes is the (n–1)th triangular number.

what comes next: 1, 3, 6, 10, 15 ..?